1.(1)資料集概述:本實驗採用 Kaggle 的惡意言論分類資料集(Jigsaw Toxic Comment Classification), 任務目標是針對網路社群留言進行多標籤分類。, 料集共包含六種互不排斥的惡意標籤toxic、severe_toxic、obscene、threat、insult以及identity_hate。單一留言可能同時觸犯多個標籤, 亦可能完全屬於正常言論(不帶有任何標籤), 此外訓練集總共有159571個留言, 測試集總共有153164個留言。

(2)初步的資料統計中, 資料集存在極其嚴重的類別不平衡(Class Imbalance)現象 。

(3)如何處理類別不平衡現象:1.用loss函數(BCEWithLogitsLoss), 可以確保資料量較少的標籤不會被其他資料量較多的標籤給蓋過, 讓像是threat也有修改權重的能力。

2.評估指標層面:Macro ROC-AUC, AUC只看模型預測後的機率, 如果預測結果不是100%XX他也有可能在這個回復中給滿分, EX:模型評估toxic機率80%正常留言20%那他還是有可能算是滿分。Marco使每個AUC算出來後加權->資料量少的標籤站有同等價值。

```
=== 各標籤的正樣本數量與比例 ===
toxic          :  15294 筆 (9.58%)
severe_toxic   :   1595 筆 (1.00%)
obscene        :   8449 筆 (5.29%)
threat         :    478 筆 (0.30%)
insult         :   7877 筆 (4.94%)
identity_hate  :   1405 筆 (0.88%)
```

2.(1)模型選擇:我使用了DistilBERT，因為它保留了BERT約 95%的語意理解能力，但參數基礎減少了40%，效率提高了許多。

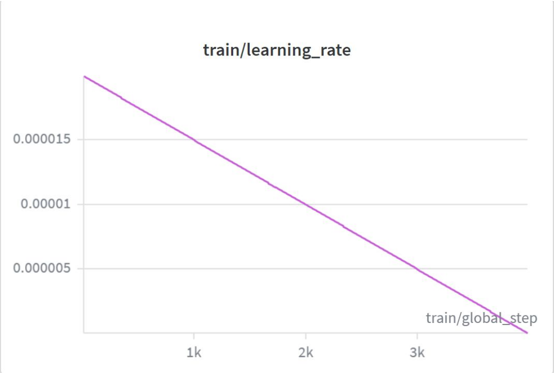
(2)整體分數對照表:

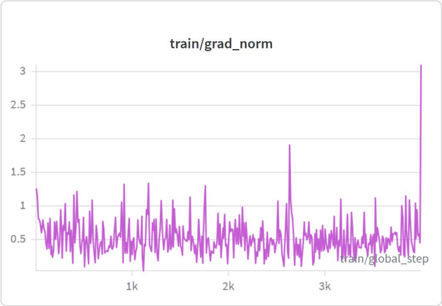
submission.csv Complete (after deadline) · 8h ago	0.98373	0.98350	

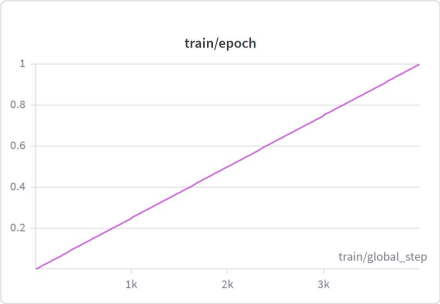
```
正在切分資料...
正在將文字轉換成 TF-IDF 向量 (這可能需要一兩分鐘，請稍候)...
開始訓練邏輯斯迴歸模型...

toxic           驗證集 AUC: 0.9662
severe_toxic    驗證集 AUC: 0.9801
obscene         驗證集 AUC: 0.9828
threat          驗證集 AUC: 0.9857
insult          驗證集 AUC: 0.9733
identity_hate   驗證集 AUC: 0.9703
------------------------------
👉 基準模型平均 AUC (Baseline Score): 0.9764
```

評估標籤 (Labels)	Baseline 驗證集 AUC	DistilBERT 驗證集 AUC	效能提升 (Up
Toxic (惡意)	0.9667	0.98807	+ 0.0213
Severe Toxic (嚴重惡意)	0.9798	0.99146	+ 0.0116
Obscene (淫穢)	0.9827	0.99447	+ 0.0117
Threat (威脅)	0.9857	0.98720	+ 0.0015
Insult (侮辱)	0.9739	0.98988	+ 0.0159
Identity Hate (身份仇恨)	0.9701	0.98704	+ 0.0169
整體平均 (Macro Average)	0.9765	0.98983	+ 0.0133
Kaggle Private LB (私有榜)	(未提交)	0.98446	—








```
正在計算驗證集成績...

🏆 各類別獨立 AUC:
- toxic: 0.98805
- severe_toxic: 0.99127
- obscene: 0.99432
- threat: 0.99160
- insult: 0.98987
- identity_hate: 0.98951

🔍 模型誤判範例:

[誤判範例 1]
📝 內容: nig, nigs? nig, nigs? what about nig, nigs? i hear sometimes people call others those. nig, nigs?...
🎯 真實標籤: [0. 0. 1. 0. 1. 1.]
🤖 模型預測: [0.722 0.011 0.082 0.024 0.23  0.058]

[誤判範例 2]
📝 內容: { { | unblock | lick my hairy nuts you negroes and jews. } }...
🎯 真實標籤: [1. 1. 0. 0. 0. 0.]
🤖 模型預測: [0.956 0.249 0.75  0.162 0.817 0.746]

[誤判範例 3]
📝 內容: " just face the truth : you can ' t escape it! = = user : arilang1234 didn ' t edit your foul, slimy racist template, you loser. i suggest that you stop pr
etending that manchus are ethnically chinese....
🎯 真實標籤: [0. 0. 1. 0. 1. 1.]
🤖 模型預測: [0.655 0.003 0.063 0.004 0.306 0.026]

[誤判範例 4]
📝 內容: n ig ger nig ger hi n i g g e r...
🎯 真實標籤: [1. 1. 1. 0. 0. 1.]
🤖 模型預測: [0.601 0.061 0.22  0.084 0.176 0.115]

[誤判範例 5]
📝 內容: " you have evidence from me! i presented you with tone of evidence, but you reject it because it goes against your fictional ideas of what new england is o
r should be. even eastern ( southeast anyway ...
🎯 真實標籤: [1. 0. 1. 0. 1. 1.]
🤖 模型預測: [0.452 0.002 0.262 0.001 0.061 0.003]
```

Submissions

0/2

You selected 0 of 2 submissions to be evaluated for your final leaderboard score. Since you selected less than 2 submissions, Kaggle auto-selected up to 2 submissions from among your public best-scoring unselected submissions for evaluation. The evaluated submission with the best Private Score is used for your final score.

■ Submissions evaluated for final score

All | Successful | Selected | Errors

Recent ▾

Submission and Description	Private Score	Public Score	Selected
submission.csv Complete (after deadline) · now	0.98373	0.98350	☐
notebookd2ee8c88cb - version 1 Complete (after deadline) · 1d ago	0.98446	0.98362	☐